IndiaKart E-Commerce Analytics Project

End-to-End Data Analysis | Phases 1 – 4

Intern Name	**Sugumar S**
Designation	**Data Analyst Intern**
Company	**Internmo**
Batch	**August 2026 – November 2026**
Project Domain	**E-Commerce Analytics**
Dataset Period	**24 June 2023 – 23 June 2025**
Records Analysed	**50,000 Orders · ~222,000 Rows · 8 Tables**
Tools Used	Python (Pandas, NumPy, Matplotlib, Seaborn) · Excel · SQL
Submission Date	**30 September 2026**

01	02	03	04
Data Cleaning	EDA & Charts	KPI Analysis	Dashboard + Report

This project was completed as part of the Data Analyst Internship at **Internmo** (Batch: **August 2026 – November 2026**). All analysis, insights, and recommendations are based on the IndiaKart dataset provided.

Sugumar S
Data Analyst Intern

1. Introduction

This report presents a complete end-to-end analytics project on IndiaKart, an Indian online marketpla Analyst Intern at Internmo (Batch: August 2026 ? November 2026), I was tasked with cleaning, exploring, analysing, a 24 months of business operations data (24 June 2023 ? 23 June 2025) covering approximately 222,000 records across 8 interconnected tables.

The goal of this project is to provide management with clear, data-driven answers to pressing business questions: which categories drive revenue, the level and patterns of nd geographies matter most, and where operational issues such as pay

2. Project Objectives

- Clean and validate the raw dataset; document data-quality issues without silent data loss.
- Perform exploratory data analysis (EDA) with 10 business-relevant visualisations.
- Calculate and interpret 10 core business KPIs against defined targets.
- Build a 5-page management dashboard and a concise written report with actionable recommendations.

3. Dataset Overview

Table	Records	Role
orders	50,000	Central order header
order_items	100,000	Line-item detail
customers	10,000	Customer profiles & segments
payments	50,000	Payment transactions
returns	10,000	Return requests & refunds
products	1,000	Product catalogue
inventory	1,000	Stock levels by warehouse
suppliers	200	Supplier master

4. Project Structure (Headings Guide)

The full project is organised into four phases. Each phase builds on the previous one.

Phase 1 — Data Cleaning & Quality Check

- Load and profile all 8 tables
- Primary-key uniqueness and referential integrity checks
- Null analysis and date-logic validation
- Business-rule checks (amounts, status consistency, COD flags)
- Issue flagging (no silent row drops) and cleaned dataset export

Phase 2 — Exploratory Data Analysis (10 Charts)

- Monthly GMV trend and seasonality
- Category-wise revenue and concentration
- Order status distribution and cancellation trend
- Customer segment value and geographic performance
- Payment methods, return reasons, and channel mix

Phase 3 — KPI Calculations

- GMV, AOV, and monthly growth rate
- Category revenue share and cancellation rate
- Payment failure rate (vs 2% target)
- Return rate, customer value, and geographic concentration
- Inventory fill rate (vs 95% target) and channel performance

Phase 4 — Dashboard & Management Report

- 5-page Excel dashboard (Scorecard, Sales, Customers, Payments, Inventory)

- Management report: Executive Summary, Top Findings, Risks, Opportunities, Recommendations
- Consolidated PDF project report for submission

5. Tools & Environment

Python (Pandas, NumPy, Matplotlib, Seaborn) for cleaning, EDA, and KPI calculation · **Jupyter / notebooks** for documented analysis · **Excel** for the interactive dashboard · **SQL schema** provided for reference.

6. Key Business Questions Addressed

? Which product categories drive the most revenue, and which underperform?
? with cancellations?
?
? Are there seasonal patterns for inventory planning?
? Which states and cities are the biggest markets?
? returns?

The detailed findings, verified KPI scorecard, charts, and recommendations follow in the main project report.

Prepared by **Sugumar S** · Data Analyst Intern · **Internmo** · Batch: **August 2026 – November 2026**

IndiaKart E-Commerce Analytics

Complete Project Report — Phases 1 to 4

Period: 24 June 2023 - 23 June 2025
Prepared: 30 September 2026 | Analy
Data: 50,000 orders · 8 tables · ~222,

1. Executive Summary

IndiaKart processed **50,000 orders** over 24 months, generating **Rs.277.91 Cr** in active GMV (**Rs.205.39 Cr** delivered). Average order value sits at **Rs.63,156**. The business is stable month-to-month with no strong growth trajectory after the initial ramp-up. Electronics dominates revenue at **57.57%**, creating meaningful concentration risk. Cancellation rate is structural at **11.99%**, of **3.50%** (above the 2% target). Return rate on eligible orders is high at **27.45%**,
. Inventory fill rate at **92.20%** is slightly below the 95% target. Uttar Pradesh and Maharashtra are the two largest markets; the App channel accounts for **55.02%** of active orders.

2. KPI Scorecard (Verified)

KPI	Value	Status
Total Orders	**50,000**	—
Active Orders	**44,004**	—
Delivered Orders	**32,499**	—
Active GMV	**Rs.277.91 Cr**	—
Delivered GMV	**Rs.205.39 Cr**	—
AOV (Active)	**Rs.63,156**	—
Cancellation Rate	**11.99%**	Watch
Payment Failure Rate	**3.50%**	Above 2% target
Return Rate (eligible)	**27.45%**	High
Inventory Fill Rate	**92.20%**	Below 95% target
Top Category Share	**57.57% Electronics**	tration
Top-5 States Share	**58.17%**	Concentrated
App Channel Share	**55.02%**	Primary

3. Top 3 Findings

1. tion
Ele el revenue (approximately Rs.145 Cr). Sports & Fitness is less than one-third that size.
A disruption or deman

2.
11.
rates are similar across methods. Cancellatio

3.
27. easons include Size Issue, Changed Mind, Damaged in Transit, Wrong Product, and Defective. Potential improvement a

4. Risks Identified

Risk	Metric	Severity
Category concentration	Electronics 57.57% of revenue	High

Payment failure above target	3.50% vs 2% target	High
Structural cancellation rate	11.99% of orders, stable	Medium-High
Inventory fill below target	92.20% vs 95%; 78 SKUs at risk	Medium

5. Opportunities

6. Recommendations (Actionable)

1. Reduce payment failures below 2%. Review payment performance by gateway and payment method. Evaluate retry logic and failure messaging, and track the failure rate weekly.

2. I
can

3. Raise inventory fill rate above 95%. Restock 76 Low Stock and 2 OOS SKUs. Tighten reorder rules for high-velocity Electronics.

4. Lower returns through size and logistics quality. Use clearer size charts for Fashion and audit packaging and
ery issues.

5. Diversify category mix over 2-3 quarters. Set a categor the contribution of Sports &
Fitness, Home & Kitchen, and Fashion to reduce depend

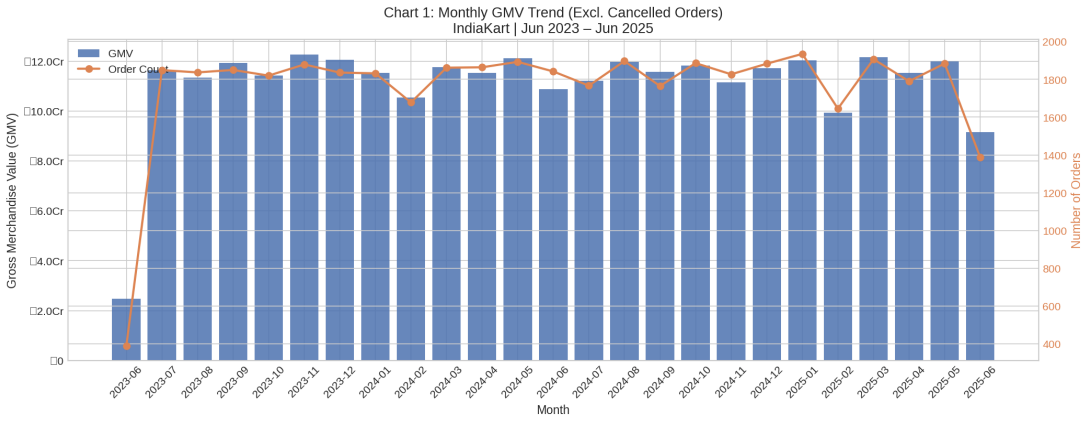
7. Project Phases Completed

Phase	Deliverable	Key Output
1. Data Cleaning	Notebook + cleaned CSVs	
2. EDA	Notebook + 10 charts	Category, geo, seasonality, returns
3. KPIs	Notebook + Excel + JSON	10 KPIs verified end-to-end
4. Dashboard	5-page Excel workbook	Scorecard, sales, customers, payments, ops
4. Report	report	Findings, risks, recommendations

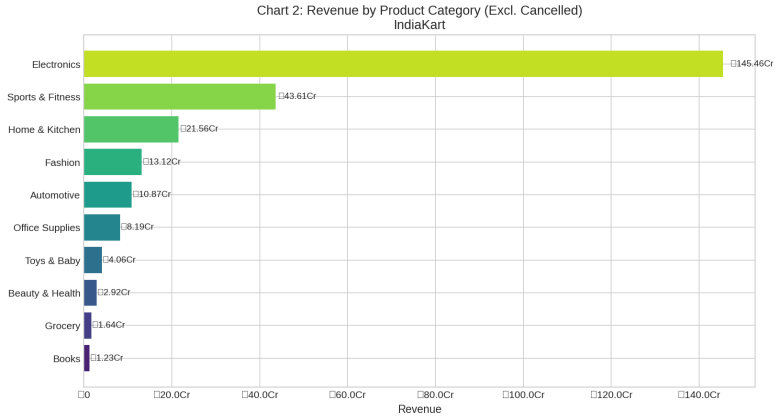
8. Data Quality Notes (Phase 1)

Primary keys unique across all 8 tables. Referential efined table relationships. delivered_date is always >= order_date (avg delivery 4.5 days). Payment amou eats (flagged, no rows dropped): 10,071 orders (20.1%) have no order_items; 10,255 orders precede registration_date (synthetic artefact); 1,740 returns linked to Cancelled/Shipped orders.

9. Key Visuals (from Phase 2 EDA)

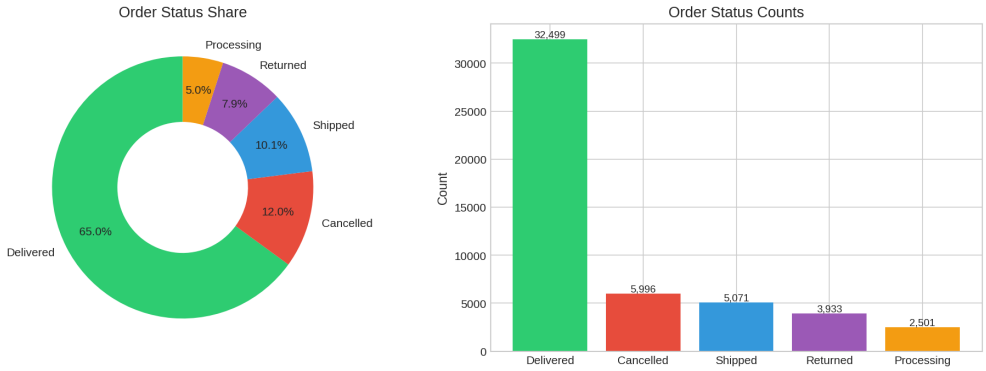


Monthly GMV Trend — stable with Nov 2023 peak

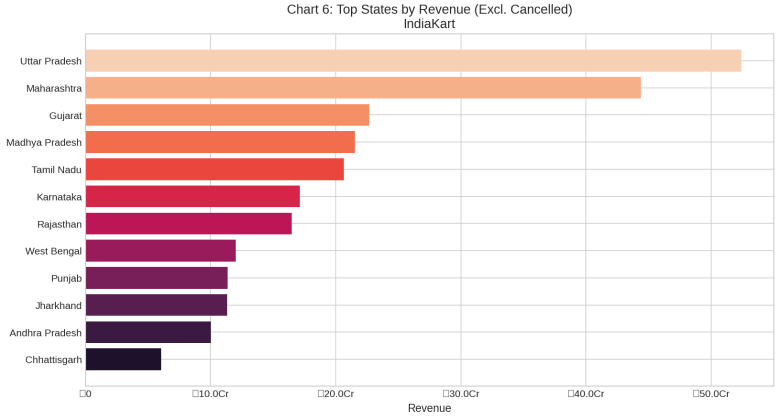


Category Revenue — Electronics dominates at 57.57%

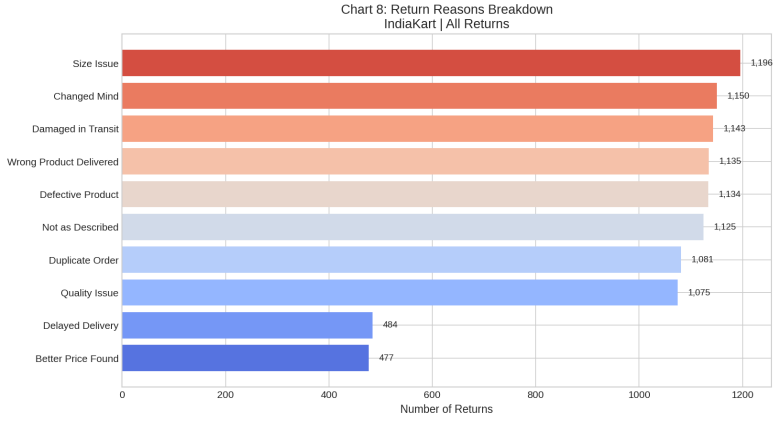
Chart 3: Order Status Distribution | IndiaKart



Order Status — 11.99% cancelled, 65% delivered

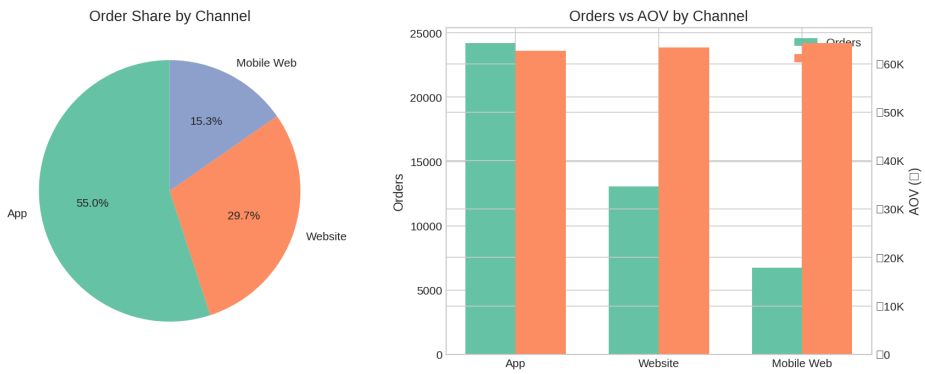


Top States — UP and Maharashtra lead



Return Reasons — size, damage, wrong item

Chart 10: Sales Channel Performance | IndiaKart



Channel Mix — App is primary at 55%

i

and significant market concentration across leading states.

(4) return quality.

This consolidated PDF contains the project findings, KPI analysis, and data-quality observations, along with visual summaries and recommendations.

- End of Report -